

Tercer Examen Parcial

$$1) y_1 = v_0 t + \frac{gt^2}{2}$$

$$y_2 = \frac{gt^2}{2}$$

$$y_1 = y_2 + 12$$

$$\frac{gt^2}{2} + 12 = v_0 t + \frac{gt^2}{2}$$

$$y_2 = \frac{9.8 \cdot 6^2}{2} = 176.4$$

$$v_0 t = 12$$

$$t = \frac{12}{20} = 0.65$$

7.76m de y_2 y a 13.76m del punto donde se arroja el primer objeto.

2)

$$t_1 = 35$$

$$v_{0y1} = 100 \sin 53^\circ$$

$$v_{01} = 100 \text{ m/s}$$

$$v_{y1} = 100 \sin 53 + 3(30 \sin 53) = 190 \sin 53$$

$$a_1 = 30 \text{ m/s}^2 \text{ a } 53^\circ$$

$$v^2 = v_0^2 + 2a_1 y$$

$$y = \frac{v_y^2 - v_{0y}^2}{2a_y} = \frac{(190 \sin 53)^2 - (100 \sin 53)^2}{2(30 \sin 53)} = 347.47$$

$$v_{y2} = 0$$

$$y = \frac{v_y^2 - v_{0y}^2}{2a_y} = \frac{0^2 - (190 \sin 53)^2}{2(-9.8)} = 1773.56 \text{ m}$$

$$v_{0y2} = 190 \sin 53$$

$$a_{y2} = 9.8 \text{ m/s}^2$$

$$y_{\max} = y_1 + y_2 = 347.47 + 1773.56 = 2121.03 \text{ m}$$

$$3) \begin{aligned} V_{rio} &= 0.500 \text{ m/s} \\ X &= 7 \text{ Km} = 7,000 \text{ m} \\ V_e &= 7.20 \text{ m/s} \end{aligned}$$

De ida =

$$\begin{aligned} V_{resultante} &= V_e - V_r \\ V_{r1} &= 7.20 - 0.500 \\ V_{r1} &= 0.7 \text{ m/s} \end{aligned} \quad \begin{aligned} t_1 &= X/V \\ t_1 &= \frac{1000 \text{ m}}{0.7 \text{ m/s}} = 1,428.57 \text{ s} \end{aligned}$$

De regreso =

$$\begin{aligned} V_{r2} &= V_e + V_r \\ V_{r2} &= 7.20 + 0.500 \\ V_{r2} &= 7.7 \text{ m/s} \end{aligned} \quad \begin{aligned} t_2 &= \frac{1000 \text{ m}}{7.7 \text{ m/s}} = 588.24 \text{ s} \end{aligned}$$

$$\begin{aligned} t_{total} &= t_1 + t_2 \\ t_{total} &= 1,428.57 + 588.24 = \underline{2,016.81 \text{ s}} \end{aligned}$$

4)

Es el mismo que el anterior

$$t_{total} = \underline{2,016.81 \text{ s}}$$

5) Eje horizontal

$$B) \sum f_x = ma = 0 = T_B \cos 45^\circ - T_A \cos 30^\circ$$
$$T_B \cos 45^\circ = T_A \cos 30^\circ$$

$$T_B = \frac{T_A \cos 30^\circ}{\cos 45^\circ}$$

$$T_B = 746.4 \text{ N} \cdot \frac{\cos 30^\circ}{\cos 45^\circ}$$

$$T_B = \underline{779.30 \text{ N}}$$

$$A) \sum f_y = ma = 0 = T_A \sin 30^\circ + T_B \sin 45^\circ - W$$
$$T_A \sin 30^\circ + \left(\frac{T_A \cos 30^\circ}{\cos 45^\circ} \right) \sin 45^\circ - W = 0$$

$$T_A \sin 30^\circ + T_A (\cos 30^\circ \tan 45^\circ) - W = 0$$

$$T_A = 0.732 W$$

$$T_A = 0.732 (200 \text{ N}) = \underline{146.4 \text{ N}}$$